Site Motion and Exact Period Counts in a Two-Layer Margolus Automaton

Route-A structural certificate C165, internal draft 0

Abstract

We study binary configurations on a ring of $2m$ sites under two staggered layers of nearest-neighbour swaps. One full tick first exchanges the even pairs and then the staggered pairs. Direct composition sends even labels two sites forward and odd labels two sites backward. By pairing each even site with a reversed odd site, we identify the configuration map with cyclic rotation of an m-letter word over a four-letter alphabet. It follows that $|\operatorname{Fix}(T^n)| = 4^{\gcd(m,n)}$, and Möbius inversion gives all exact-period configurations and primitive cycles. These are all-size source theorems, not extrapolations from the finite checks. No target divisor, arithmetic factor, or spectral operator claim is made.

Keywords: partitioned cellular automaton; Margolus blocks; cyclic rotation; necklace; exact period; primitive cycle; dynamical zeta.

中文摘要

本文研究 $2m$ 个站点的二元环，其中一个完整时钟依次执行两层交错近邻交换。 直接跟踪带标签的元胞可知，偶站点向前移动两格，奇站点向后移动两格。 把偶站点与反向编号的奇站点配对后，整个配置动力系统严格等价于四字母长度 m 词的循环旋转。因此可以得到所有时刻的不动配置数，并由莫比乌斯反演恢复 精确周期配置和本原周期。有限枚举仅用于回归检查，不承担全参数定理的证明。 本文不比较任何目标除子，也不提出算术因子或谱算子结论。

关键词：分块元胞自动机；交错交换；循环旋转；项链；精确周期； 本原周期；动力 ζ 函数。

1 Full-tick site law

Let $X_m = \{0,1\}^{\mathbb{Z}/(2m)\mathbb{Z}}$. Layer A swaps $(0,1),(2,3),\ldots$, while B swaps $(1,2),(3,4),\ldots,(2m-1,0)$. One tick is $T = B \circ A$. The labelled-cell permutation is

$$\tau(i) = \begin{cases} i+2, & i \text{ even}, \\ i-2, & i \text{ odd}, \end{cases} \pmod{2m}. \tag{1}$$

Put $e_j = 2j$ and $o_j = 1-2j$. Then $\tau(e_j) = e_{j+1}$ and $\tau(o_j) = o_{j+1}$. Hence

$$\Phi(x)_j = (x_{e_j}, x_{o_j}) \in \{0,1\}^2 \tag{2}$$

conjugates T to cyclic rotation of a four-letter word of length m.

2 Period law

The nth power of the rotation has $\gcd(m,n)$ coordinate cycles, proving

$$|\operatorname{Fix}(T^n)| = 4^{\gcd(m,n)}. \tag{3}$$

Every exact period divides m. For $d \mid m$,

$$P_m(d) = \sum_{e \mid d} \mu(d/e) 4^e, \qquad C_m(d) = P_m(d)/d. \tag{4}$$

Thus the finite source zeta is

$$\zeta_T(z) = \prod_{d \mid m} (1-z^d)^{-C_m(d)}. \tag{5}$$

At $m=1$ the full tick is the identity on four configurations. At $m=2$ there are four fixed configurations and twelve configurations in six two-cycles.

Exact ledgers through $m=16$ and direct enumeration through $m=8$ are regression sentinels. The strict preliminary tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT). No target table, Euler factor, root number, automorphy, Hilbert–Pólya operator, or Route-B input is used. Scope:

NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data and code availability. Package-local exact evidence and code accompany the manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Conflicts of interest. None known.

Funding. No external funding is reported.

AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer.